DESIGN OF ELECTRONIC VOTING MACHINE

By

AJAY P (Reg. no. 9517202403007)
VIJAY SRIDHAR P (Reg. no. 9517202403181)

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CERTIFICATE

This is to certify that it is the Bonafide work done by Mr. AJAY P (9517202403007), Mr. VIJAY SRIDHAR P (9517202403181) for the 23EC353 - Mini Project - I work entitled Electronic Voting Machine using CD4026 at Department of Electronics and Communication Engineering, Mepco Schlenk Engineering College, Sivakasi during the year 2025-2026.

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Faculty In-Charge ***Head of the Department***

Submitted for the Practical Examination held at Mepco Schlenk Engineering College, Sivakasi on ……………………….

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Internal Examiner - I ***InternalExaminer - II***

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ABSTRACT

The mini project titled **"Electronic Voting Machine Using IC CD4026"** presents a simple, reliable, and cost-effective prototype designed to demonstrate the fundamental working principles of an electronic voting system.The system employs the CD4026 IC—an integrated decade counter with a decoded seven-segment display output—which plays a crucial role in counting and visually presenting the number of votes cast for each candidate.

In this design, each candidate is assigned a dedicated push-button switch that, when pressed, triggers the CD4026 counter to increment by one. The IC then drives a seven-segment display to show the updated vote count in real time. To ensure transparency and accuracy, debouncing methods are incorporated to minimize false triggers due to mechanical switch noise.

Although it does not incorporate advanced security features such as authentication or data encryption, it successfully demonstrates the underlying principle of electronic vote tallying and real-time result visualization. Overall, the Electronic Voting Machine using the CD4026 IC serves as an excellent educational tool that bridges theoretical digital electronics concepts with practical implementation

1. INTRODUCTION

ELECTRONIC VOTING MACHINE-EVM's

Electronic Voting Machines (EVMs) represent a significant technological advancement in the process of conducting elections. Designed to replace traditional paper-based voting systems, aim to EVMs enhance the efficiency, accuracy, and reliability of the voting and vote-counting process. At their core, EVMs are electronic devices that allow voters to cast their ballots through a simple interface, typically involving a control unit operated by election officials and a balloting unit used by voters. This digital process not only streamlines the overall conduct of elections but also minimizes many common issues associated with manual voting methods.

Historically, paper ballots were prone to errors, manipulation, invalid votes, and lengthy counting procedures. These challenges often resulted in delays, disputes, and high administrative costs. Electronic Voting Machines were developed to overcome such limitations by ensuring greater security, reducing human intervention, and promoting transparency. Through features such as one vote per activation, secure internal memory, and tamper-resistant hardware, EVMs have become a trusted tool in many electoral systems around the world. They have been especially valuable in regions with large populations, where rapid counting and error reduction are essential.

DIFFERENT TYPES OF EVM's

Electronic Voting Machines have evolved over time, and several types are used worldwide depending on technological needs, security requirements, and the scale of elections. One common type is the Direct Recording Electronic (DRE) EVM, where voters cast their ballots using a touchscreen or push-button interface, and votes are recorded directly into electronic memory. Some DRE systems include a Voter Verifiable Paper Audit Trail (VVPAT), which provides a printed record for verification and auditing.

Another widely used type is the **Ballot Unit and Control Unit–based EVM**, commonly deployed in countries like India. This system uses a simple push-button interface on the Ballot Unit connected to a Control Unit that securely stores vote data.

More advanced systems include **optical scan voting machines**, which allow voters to mark paper ballots that are then scanned electronically for counting. There are also **internet-based or remote voting systems**, still emerging, which enable voting over secure digital platforms for specific cases such as overseas or disabled voters.

ADVANTGES

- Speed and Efficiency.
- Reduced Human Error.
- No need for Paper Ballots.

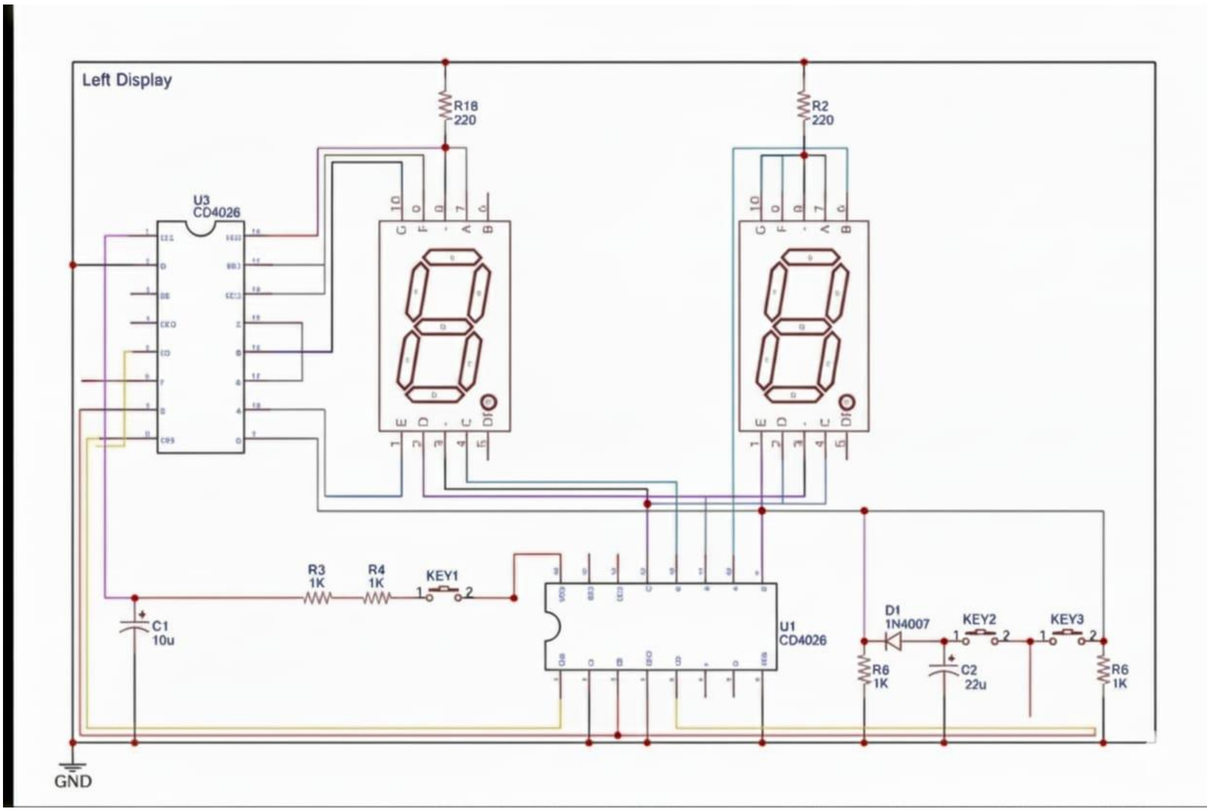
DISADVANTAGES

- Vulnerability / Tampering.
- Lack of Paper Trail.
- Voter Confidence Issues.

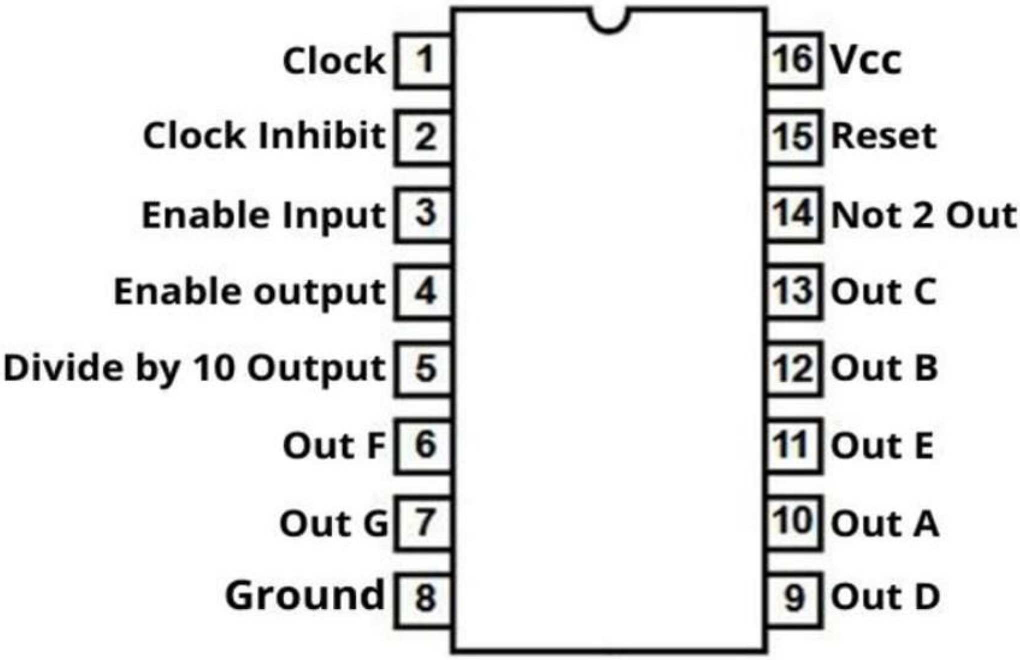
APPLICATIONS

- National and Regional Elections: Widely used in general elections to record votes
  electronically, ensuring faster and more accurate counting of votes.
- Polling for Referendums: Used to conduct referendums, enabling quick vote tallying and transparent results.
- By-elections: Deployed to fill vacant legislative or parliamentary seats between general elections, improving efficiency and maintaining electoral integrity.
- Corporate or Organizational Voting: Employed in private sector elections, such as for board appointments or shareholder voting, ensuring security and confidentiality.
- E-Voting for Overseas Citizens: Facilitates voting for citizens living abroad, using secure remote voting methods to participate in national elections.

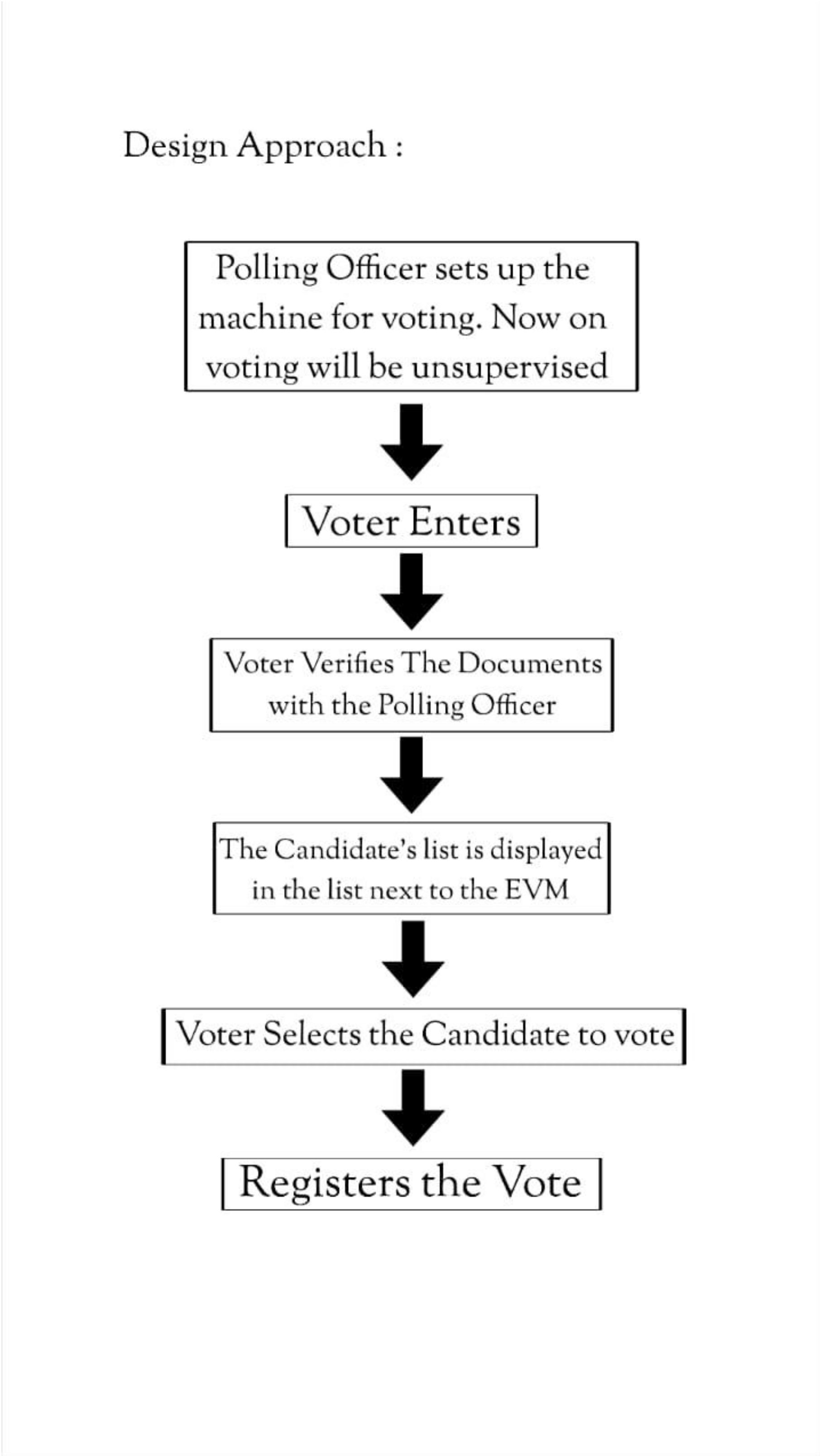
APPLIED CIRCUIT



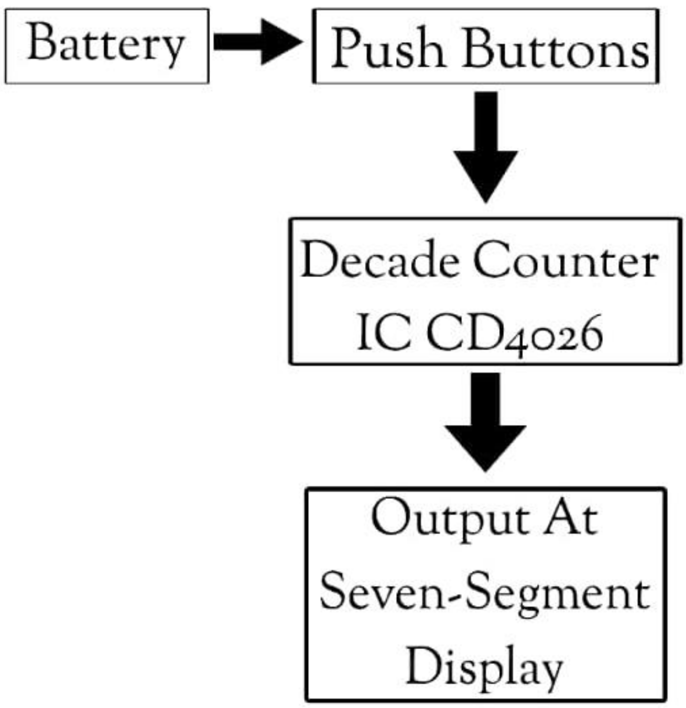
IC CD4026-PIN DIAGRAM



DESIGN APPROACH



BLOCK SCHEMATIC



CIRCUIT WORKING BLOCKS AND TESTING

1.CD4026 as Counter + 7-Segment Driver

The CD4026 IC receives the pulse from the push button and increments its internal counter.
It directly drives the 7-segment display, showing the updated vote count (0–9 per IC, or 00–99 if cascaded).

2. Separate Counting Module for Each Candidate

Each candidate's votes are counted independently using:
-One push button
-One CD4026 IC
-One 7-segment display (or a pair for two-digit count)
This prevents any cross-mixing of votes.

3. Power Supply Using a 9V Battery

A 9V battery powers the breadboard modules.
Voltage is regulated through resistors to protect the CD4026 and displays.

4. Debouncing Through Resistors / RC Network

Mechanical switches produce noise (bouncing).
Your circuit uses resistors and wiring arrangements to reduce false multiple counts caused by switch bounce.

5. Output Display Through 7-Segment Displays

The votes are shown directly on the 7-segment displays.
As seen in your images, the displays show values like 21, 04, etc.

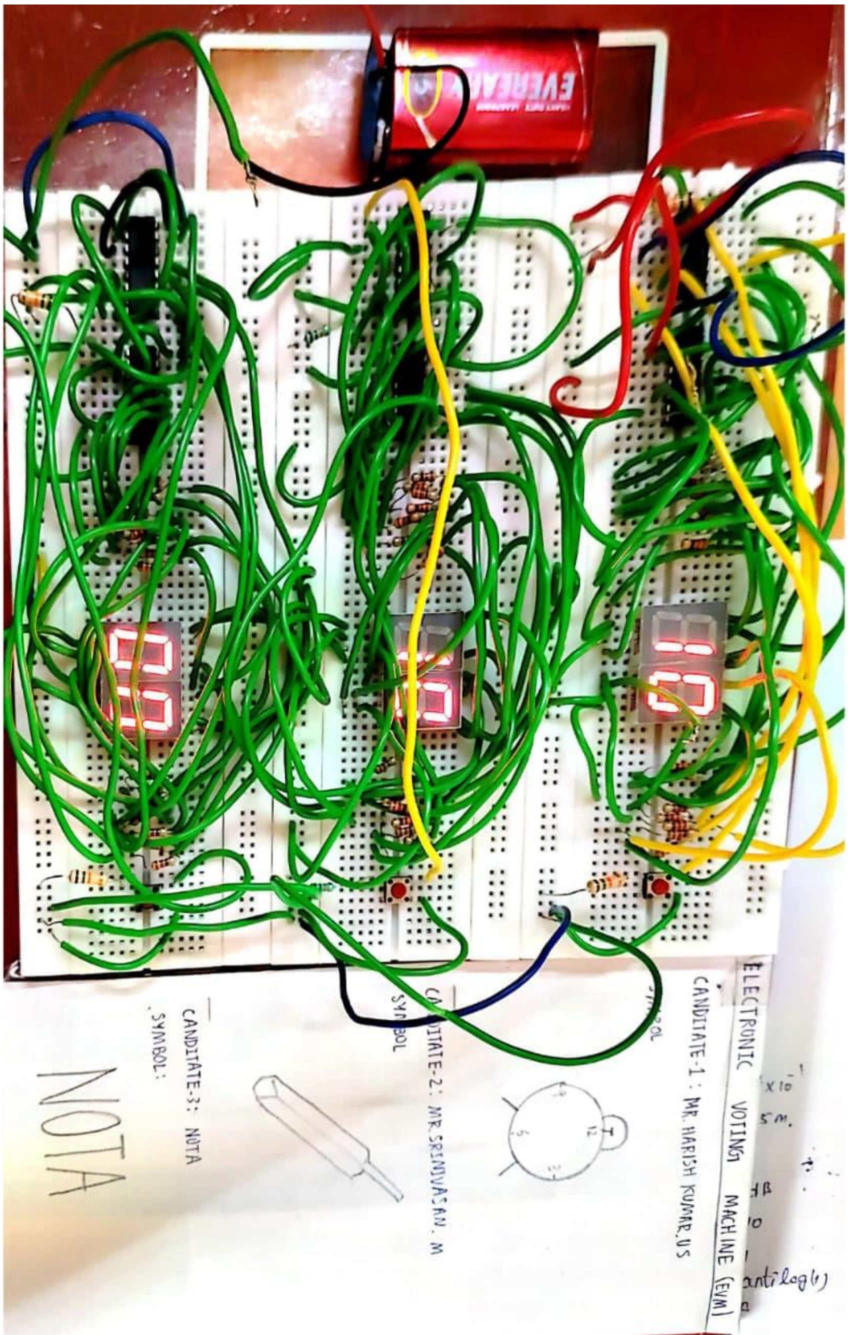
6. Wiring for Multi-Digit Display

To count above 9, two CD4026 ICs are cascaded:
-First IC handles units place
-Second IC handles tens place
-Carry-out output of IC1 goes into clock input of IC2.

CIRCUIT CONNECTIONS

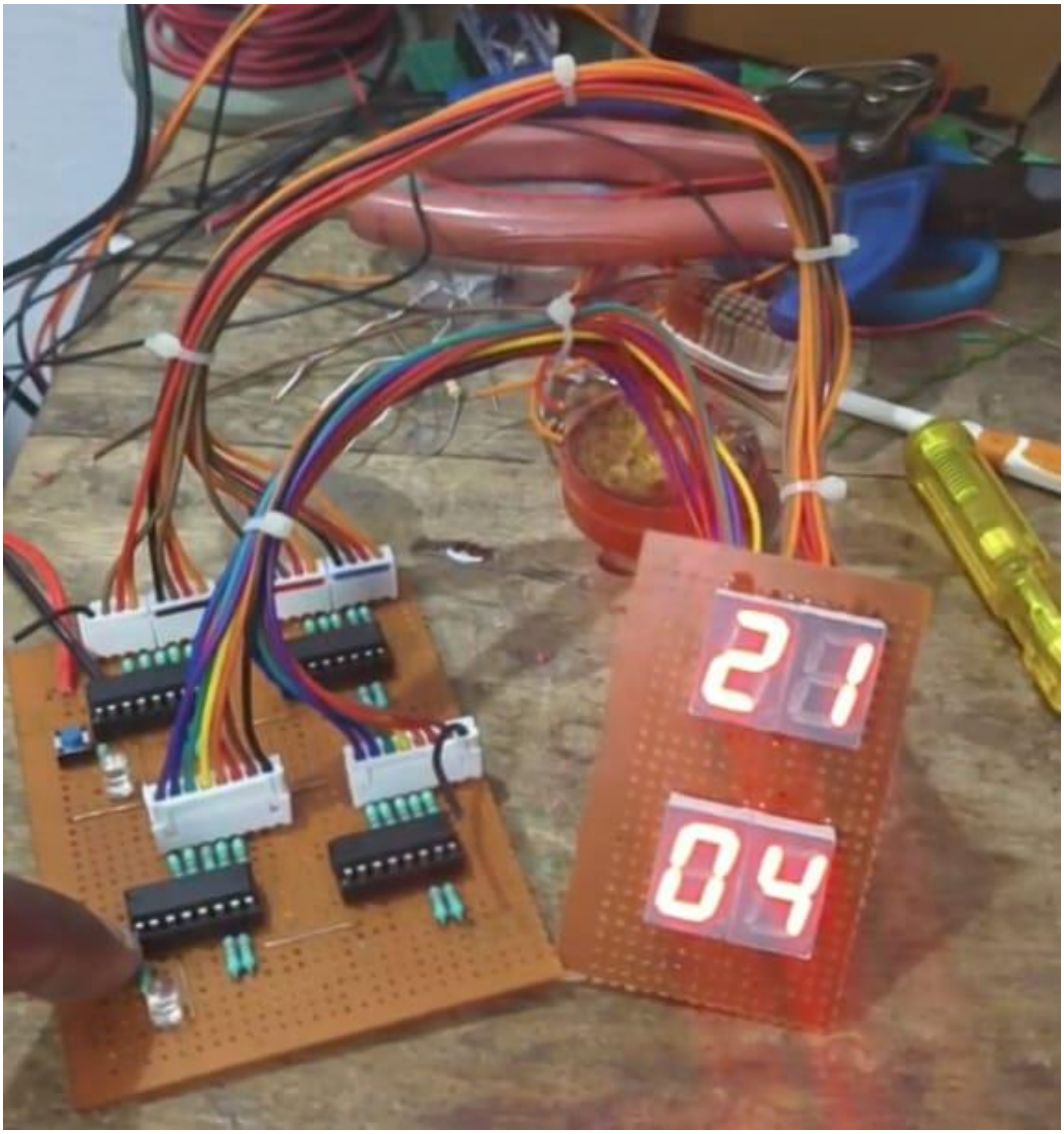
1.BREABBOARD IMPLEMENTATION

The entire circuit is built on interconnected breadboards, using multiple IC CD4026 counters, seven-segment displays, and push-button inputs. The layout demonstrates a fully functional voting system prototype assembled purely through breadboard wiring and discrete components.



2.PCB IMPLEMENTATION

The entire circuit is neatly designed on a printed circuit board, providing a more stable and permanent setup compared to the breadboard version.Each section of the PCB is dedicated to a candidate, with IC CD4026 counters, seven-segment displays, and switches precisely soldered in place.The PCB layout demonstrates an organized, professional, and efficient version of the voting system, suitable for long-term use and testing.



CURRENT EVM USED IN INDIA

The Electronic Voting Machine (EVM) used in India is a reliable and tamper-proof device designed by the Election Commission of India to ensure secure and transparent voting. It consists of two main units: the Ballot Unit and the Control Unit, connected through a cable. The Ballot Unit contains candidate names and symbols, allowing voters to cast their vote by pressing a single button. The Control Unit is operated only by authorized polling officers and records each vote securely. Indian EVMs are stand-alone, battery-operated devices with no internet or wireless connectivity, eliminating hacking risks. They use advanced microcontrollers and encrypted data storage for high security. Each vote is stored electronically and can be verified using VVPAT, which provides a printed slip visible to the voter for a few seconds. These machines support multi-candidate elections and ensure quick, error-free counting. The EVM system has improved efficiency, reduced invalid votes, and strengthened the integrity of the electoral process in India.

CONCLUSION

In conclusion, Electronic Voting Machines (EVMs) represent a transformative step forward in the modernization of electoral processes. They offer significant improvements over traditional paper-based voting systems by ensuring faster, more accurate vote recording, reducing human error, and minimizing the risk of fraudulent activities. EVMs have not only enhanced the efficiency and security of elections but have also made the electoral process more transparent and accessible to voters

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